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Comparison of Direct and Cascaded High-Resolution Spatial Downscaling of Wildfire Data

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Code Repository:
https://github.com/ese-ada-lovelace-2025/irp-th25

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Abstract

With climate change, there's an increasing need for a higher-resolution wildfire data for better monitoring and risk assessment. GEO satellites have high-temporal resolution, but lacks spatial resolution for a detailed risk assessment. While LEO satellites provide better spatial resolution, but low temporal resolution to monitor rapid wildfire spread. Existing downscaling models have demonstrated their effectiveness in producing high-resolution remote sensing data. But many still fail to capture the fine details and physically plausible outputs. This project aims to achieve a high-resolution spatial downscaling of wildfire data by integrating additional physical covariates to the model to mitigate implausible outputs. The project explores direct (2km to 30m) and cascaded (2km to 375m to 30m) downscaling pipelines and their effectiveness in producing high-resolution wildfire data. The project compares diffusion (EDiffSR), attention U-Net (HAUNet), and transformer-based (ESTNet) downscaling models.

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1 Introduction

In recent years, due to rising popularity in Net-Zero initiatives, carbon investors are increasingly funding carbon reduction and removal projects, such as reforestation (Lambert and Turner, 2024). However, these carbon projects are at risk because wildfires can destroy and release the carbon back into the atmosphere. With climate change, wildfires are becoming increasingly severe and frequent (Jones et al., 2020). This causes economic losses for the insurance industry. For instance, The January 2025 Los Angeles County wildfires set an all-time record, causing $40 billion in insured losses alone from a single event (Dwyer and Nam, 2026). This project, conducted with Artio, aims to address this gap. By providing highly accurate risk assessment models, this research seeks to give better confidence and enable better risk assessment.

High-spatial and high-temporal resolution wildfire data is favourable in risk assessment. However, high-temporal resolution data obtained from geostationary satellites (GEO) have limited spatial resolution, which makes them difficult to use for accurate risk assessment. While low orbit satellites (LEO) offers high-spatial resolution, but low-temporal resolution.

To bridge this gap, existing methods uses Deep Learning (DL) models for downscaling due to their ability in recognising non-linear spatiotemporal patterns. However, the resolution gap between GEO and LEO is significant (2km - 30m). This increases the risk of producing physically implausible or unrealistic outputs, as a single low-resolution (LR) pixel can correspond to different high-resolution (HR) details (Z. Wang et al., 2020). In this project, to mitigate this limitation, the model integrates additional physical information, such as HR topography and landcover to help guide the downscaling process. To further improve downscaling, the project aims to investigate whether cascaded downscaling (2km to 375m to 30m) produces more accurate and better results compared to direct downscaling (2km to 30m). By breaking up the downscaling process, the model trains on a smaller resolution gap, aiming to mitigate physically implausible outputs. This project will compare the cascaded and direct pipeline performance of different downscaling architectures, including hybrid attention U-Net, a transformer-based model, and a diffusion-based model.

2 Current State of Remote Sensing Downscaling

Traditional downscaling methods often fail to capture the complex non-linear patterns and variability in remote sensing data. In recent years, DL methods have shown promising results that outpaces traditional approaches (Sdraka et al., 2022). In particular, Convolutional Neural Networks (CNNs) and U-Net architectures have been widely used for their ability to capture spatial features. Generative Adversarial Networks (GANs) have also been used for their ability to generate sharp images. Recently, Transformer and Diffusion-based approaches have been proposed for downscaling. Transformers use attention mechanisms to capture long-range dependencies (J. Wang et al., 2025), while Diffusion models shows good generative ability and fidelity (Y. Wang et al., 2023).

However, many of these approaches struggles preserving fine textures and geometric structures (Qi et al., 2026). This is problematic for wildfire data, where the spatial patterns and textures are crucial for risk assessment. To address these limitations, there is a need for domain-specific architectures that can better capture the characteristics of wildfire data.

Existing literature remains reliant on direct downscaling methods across large resolution-gaps. Some studies have explored cascaded approaches, such as AMCN-Geoi-DBN from Jing et al., 2022, and CLIT from Chen et al., 2023, but there are currently no research from the literature investigating cascaded downscaling in the context of wildfires. The limitations and benefits of cascaded downscaling is also largely unexplored. To address this gap, the project objectives are:

1. Develop a domain-specific, multi-stage framework for wildfire downsclaing.

2. Compare the performance of different downscaling architectures in terms of preserving fine textures and geometric structures.

3. Evaluate the trade-offs of cascaded and direct downscaling and its effectiveness in improving the accuracy of wildfire downscaling.

3 Data and Materials

3.1 Input Datasets

- GOES-17 channel 7 (IR shortwave) of Level 1B (L1B) Radiances product (ABI-L1B-Rad). Has spatial resolution of 2km at every 5 mins. This channel is selected because it is commonly used to identify hot regions (Badhan et al., 2024). This will be the main coarse input.

- Copernicus DEM GLO-30: Global 30m Digital Elevation Model with 30m resolution (European Space Agency and Airbus, 2022). DEM band will be used.

- ESA WorldCover 10m v100 landcover dataset with 10m resolution. This dataset have 11 land cover classes, including tree cover, shrubland, grassland, and permanent water bodies (Zanaga et al., 2021).

3.2 Target Fire Data

- Landsat-8 active fire dataset includes data for fire scenes taken from August and September 2020 with dimensions 256x256 at 30m resolution (de Almeida Pereira et al., 2021). This will be the final target HR target data. Fire-sensitive SWIR and thermal bands will be used.

- VIIRS (S-NPP) Band Active Fire with 375 m spatial resolution (Earth Science Data Systems, 2025) will be used as the target data for the 2km to 375m stage downscaling. The I-4 channel brightness temperature (3.55-3.93 µm) band will be used.

4 Methods

4.1 HAUNet (Hybrid Attention U-Net)

Among CNN-based approaches, HAUNet (J. Wang et al., 2023) uses hybrid attention to capture both local fine detail and broader contextual structure across multiple scales. HAUNet was selected for its ability to capture local extreme hotspots in near-zero background, which standard U-Nets might smooth out. HAUNet uses two variants of attention blocks:

- **CEM**: Applies channel attention.

- **S_CEM**: Additionally applies spatial attention in parallel.

The model uses cross-scale attention module as skip connections across different scale outputs in the encoder, to capture dependency between different scales. The overall framework of wildfire-adapted HAUNet is shown in Figure 1. The original HAUNet S_CEM does full spatial attention, which is $O((H \cdot W)^2)$ and will run into OOM crashes. Hence the input GOES and covariates are sampled to a smaller image before feeding into the model. But this loses covariates' high-resolution information, so an additional covariate pyramid encoder was added form the original model to run in parallel. This performs windowed spatial attention on the covariates at fine resolution. The output of the covariate encoders are fused at different scales during refinement.

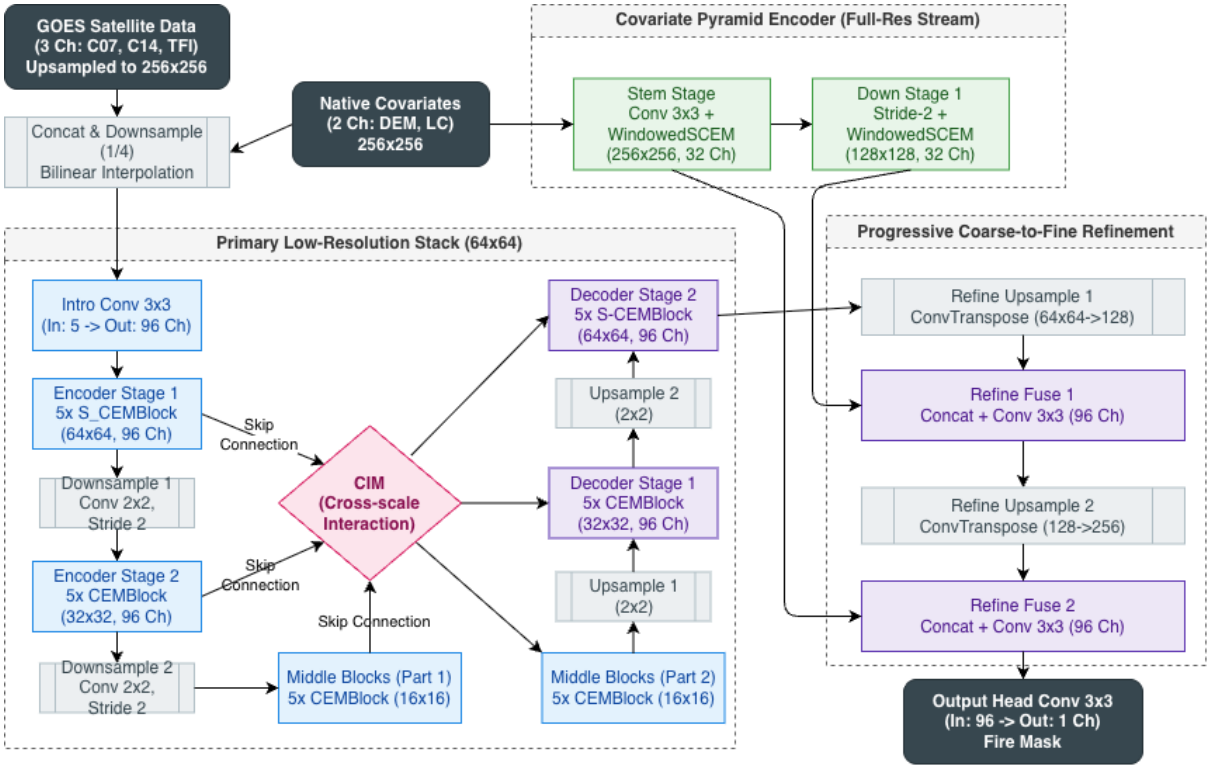


Figure 1: Architecture overview of the wildfire-adapted HAUNet. The full-resolution stream processes native covariates, which are progressively fused via refinement blocks with the low-resolution stack after CIM.

5 Results

6 Discussions

7 Conclusions

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